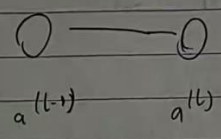
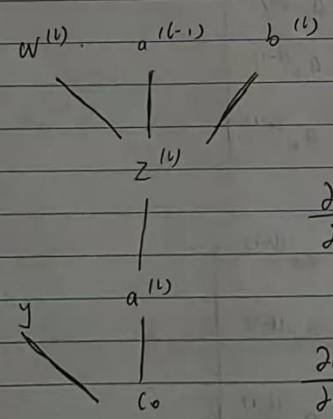
Relu Relu e → 支付



$(a^{(l)} - y)^2 = \text{const}$

$z^{(l)} = w^{(l)} a^{(l-1)} + b^{(l)}$ $\quad a^{(l)} = \sigma(z^{(l)})$



$$\frac{\partial C_0}{\partial w^{(l)}} = \frac{\partial z^{(l)}}{\partial w^{(l)}} \cdot \frac{\partial a^{(l)}}{\partial z^{(l)}} \cdot \frac{\partial C_0}{\partial a^{(l)}}$$

$$= 2(a^{(l)} - y) \cdot \sigma'(z^{(l)}) \cdot a^{(l-1)}$$

$$\frac{\partial C}{\partial w^{(l)}} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial w^{(l)}}$$

$$\nabla C = \begin{pmatrix} \frac{\partial C}{\partial w^{(1)}} \\ \frac{\partial C}{\partial b^{(1)}} \\ \vdots \\ \frac{\partial C}{\partial w^{(l)}} \\ \vdots \\ \frac{\partial C}{\partial b^{(l)}} \end{pmatrix}$$

$$\frac{\partial C_0}{\partial b^{(l)}} = \frac{\partial z^{(l)}}{\partial b^{(l)}} \cdot \frac{\partial a^{(l)}}{\partial z^{(l)}} \cdot \frac{\partial C_0}{\partial a^{(l)}}$$

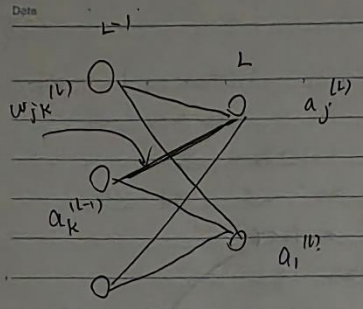
$$C = \left[\sigma(w^{(l)} a^{(l-1)} + b^{(l)}) - y\right]^2$$

$$\frac{\partial C_0}{\partial b^{(l)}} = \frac{\partial C_0}{\partial a^{(l)}} \cdot \frac{\partial a^{(l)}}{\partial z^{(l)}} \cdot \frac{\partial z^{(l)}}{\partial b^{(l)}}$$

$$= 2(a^{(l)} - y) \cdot \sigma'(z^{(l)}) \cdot 1$$

No.

Date



$$C_0 = (a_0^{(L)} - y_0)^2 + (a_1^{(L)} - y_1)^2$$

$$= \sum_{j=0}^{n_L-1} (a_j^{(L)} - y_j)^2$$

$$z_j^{(L)} = w_{j_0}^{(L)} a_0^{(L-1)} + w_{j_1}^{(L)} a_1^{(L-1)} + w_{j_2}^{(L)} a_2^{(L-1)} + b_j^{(L)}$$

$$\underbrace{\begin{bmatrix} w_{j_0}^{(L)} & w_{j_1}^{(L)} & w_{j_2}^{(L)} \end{bmatrix}}_{W} \begin{bmatrix} a_0^{(L-1)} \\ a_1^{(L-1)} \\ a_2^{(L-1)} \end{bmatrix}$$

$$\begin{bmatrix} W_{00}^{(L)} & W_{01}^{(L)} & W_{02}^{(L)} \\ W_{10}^{(L)} & W_{11}^{(L)} & W_{12}^{(L)} \end{bmatrix} \begin{bmatrix} a_0^{(L-1)} \\ a_1^{(L-1)} \\ a_2^{(L-1)} \end{bmatrix}$$

$$= \begin{bmatrix} z_0^{(L)} \\ z_1^{(L)} \end{bmatrix}$$

$$z_0^{(L)} = W_{00}^{(L)} a_0^{(L-1)} + W_{01}^{(L)} a_1^{(L-1)} + W_{02}^{(L)} a_2^{(L-1)} + b_0^{(L)}$$

$$z_1^{(L)} = W_{10}^{(L)} a_0^{(L-1)} + W_{11}^{(L)} a_1^{(L-1)} + W_{12}^{(L)} a_2^{(L-1)} + b_1^{(L)}$$

$$a_0^{(L)} = \sigma(z_0^{(L)}) \qquad C_0 = \sum (a_j^{(L)} - y)^2$$

C_0 $\quad W_{00}^{(L)}$ $\quad W_{01}^{(L)}$ $\quad W_{02}^{(L)}$ $\quad a_0^{(L-1)}$ $\quad a_1^{(L-1)}$ $\quad a_2^{(L-1)}$ $\quad b_0^{(L)}$

$$\frac{\partial C_0}{\partial W_{00}^{(L)}} = 2(a_0^{(L)} - y)\,\sigma'(z_0^{(L)})\, a_0^{(L-1)} \qquad \frac{\partial C_0}{\partial a_0^{(L-1)}} = 2(a_0^{(L)} - y)\,\sigma'(z_0^{(L)})\, W_{00}^{(L)}$$

No.

Date

$$C_0 \begin{cases} a_0 \longrightarrow \sigma(z_0) \rightarrow a_0^{(l-1)} \\ a_1 \longrightarrow \sigma(z_1) \rightarrow a_0^{(l-1)} \end{cases}$$

$$\frac{\partial C_0}{\partial a_0^{(l-1)}} = 2(a_0^{(l)} - y)\sigma'(z_0^{(l)}) W_{00}^{(l)} + 2(a_1^{(l)} - y)\sigma'(z_1^{(l)}) W_{10}^{(l)}$$

$$= \sum_{j=0}^{n_{l-1}} \frac{\partial z_j^{(l)}}{a_k^{(l-1)}} \cdot \frac{\partial a_j^{(l)}}{\partial z_j^{(l)}} \cdot \frac{\partial C_0}{\partial a_j^{(l)}}$$

$a_0^{(l)} \uparrow \quad b_0^{(l)} \uparrow \quad w^{(l)} \uparrow \quad a^{(l-1)} \uparrow$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\cancel{\sigma'(z) = -\frac{1}{(1+e^{-z})^2}(1-e^{-z}) = -\frac{1}{1+e^{-z}} \cdot \frac{1}{1+e^{-z}}(1-e^{-z})}$$

$$\sigma'(z) = -\frac{1}{(1+e^{-z})^2}(-e^{-z}) = \cancel{-\frac{1}{1+e^{-z}} \cdot \frac{1+e^{-z}-2e^{-z}}{1+e^{-z}}}$$

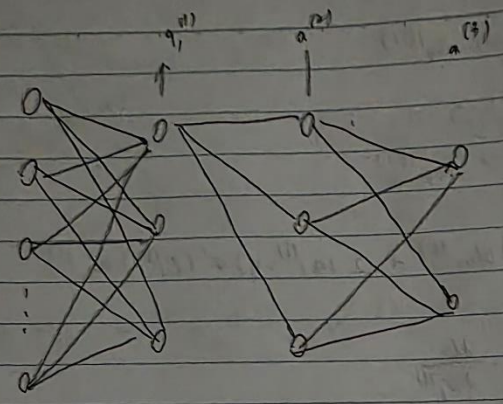
$$= \frac{e^{-z}}{(1+e^{-z})^2} = \cancel{-\frac{1}{1+e^{-z}} \cdot \left(1 - \frac{2e^{-z}}{1+e^{-z}}\right)}$$

$$= \frac{1+e^{-z}-1}{(1+e^{-z})^2} = \cancel{-\frac{1}{1+e^{-z}} \cdot \left(1 - \frac{2}{e^{z}+1}\right)}$$

$$= \frac{1}{1+e^{-z}} - \frac{1}{(1+e^{-z})^2} = \cancel{\quad}$$

$$= \sigma(z) - \sigma(z)^2 \qquad a = \sigma(z)$$

$$= \sigma(z)(1-\sigma(z)) \qquad a(1-a)$$



$|a^{(L)} - y|^2 = C \qquad \frac{\partial C}{\partial a_L} = 2(a^{(L)} - y)$

$\sigma'(L) = L(1-L)$

$\begin{bmatrix} \\ \end{bmatrix} \leftarrow 2(a_3 - y) \; * \begin{bmatrix} \\ \end{bmatrix} \Rightarrow \begin{bmatrix} \\ \end{bmatrix}$

$\begin{vmatrix} 1 \\ 2 \end{vmatrix} \begin{vmatrix} 1 & 3 & 4 & 1 \end{vmatrix} = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}$

$\begin{bmatrix} \\ \end{bmatrix} \begin{pmatrix} \quad \end{pmatrix} = 3 \begin{bmatrix} \\ \end{bmatrix}$

$(1 \times 2)(2 \times 3)$

$(2 \times 1)(1 \times 3) \rightarrow (2 \times 3) \qquad W_3 = W_3 - \square$